

Collagen Fibre Orientation Disorder as a Prognostic Biomarker, and an Agentic Framework for Histopathology Tool Orchestration

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Abstract. This report documents two projects with deliberately opposite characters. The first extended prior work on collagen fibre orientation disorder (CFOD) from feature extraction into survival analysis across the TaCTIL cohort, 816 breast cancer patients from five institutions, under progressively stricter validation frameworks. Of the three feature families tested, CFOD_G was the most consistently prognostic, but the majority of results did not reach significance at $\alpha = 0.05$, and changing the endpoint from disease-free to overall survival weakened them further. That negative result, and specifically the magnitude of the shift caused by the endpoint change alone, is the substantive finding. The second project built an agentic orchestration layer over the lab's existing feature extraction pipelines, allowing a pathologist to express an intent in natural language and have the system select, execute and explain only the relevant tools. Both projects are reported here with methods, results and the technical obstacles encountered.

Keywords. computational pathology; collagen fibre orientation disorder; survival analysis; multi-institutional validation; agentic tool orchestration

1 Project I: CFOD features and survival analysis

1.1 Background and objective

Collagen fibre orientation disorder has become one of the more promising candidate biomarkers in cancer research, because it quantifies the interaction between a tumour and the surrounding stromal microenvironment rather than the tumour alone. Prior work in the lab established the feature extraction pipeline. The objective this semester was narrower and harder: determine whether those features carry prognostic information about patient outcome when validated across institutions, rather than whether they can be computed reliably.

1.2 Cohort and feature families

Analysis used the TaCTIL dataset, 816 patients drawn from five institutions (Table 1). Three CFOD-derived feature families were evaluated: CFOD, CFOD_G and COLA. Each was assessed at both median and 80th-percentile risk stratification thresholds, with significance judged at $\alpha = 0.05$.

1.3 Validation frameworks

Three frameworks were applied in sequence, each stricter than the last.

1. **Single-institution holdout.** Models were trained on data from one institution and validated on the remaining four. Training on Washington University data and testing on the other four is the representative case. This tests generalisation across patient populations directly, at the cost of a small training set.

Table 1. TaCTIL cohort composition.

Institution	Patients
Cleveland Clinic Foundation (CCF)	336
Vanderbilt University	158
Johns Hopkins University (JHU)	121
Washington University (WU)	107
Houston	94
Total	816

2. **Endpoint change.** Midway through the semester, after consultation with collaborators, the endpoint was changed from disease-free survival (DFS) to overall survival (OS). This required substituting the outcome and time-to-event fields throughout and re-running every prior analysis.
3. **Structured development and testing split.** With guidance from German, a member of the lab, a formal split was implemented: approximately 44 % of patients (360) in development and 56 % (456) in testing. Within development, train-to-validation ratios of 60:40 and 70:30 were both trialled before evaluation on the held-out test set. Because the test set was larger than the development set, the two were later swapped to check whether a larger development set recovered any signal.

1.4 Results

The representative significant result is a Kaplan–Meier separation on a holdout set of $N = 580$, with $p = 3.39 \times 10^{-2}$ and a hazard ratio of 2.57 (95 % CI 0.30–21.78). The high-risk curve sits consistently below the low-risk curve and $p < 0.05$, which is the pattern the analysis was looking for. The confidence interval on the hazard ratio spans unity by a wide margin, which should temper any reading of that result as strong.

Table 2. Single-institution holdout results under the original DFS endpoint. Cells name the feature family reaching significance; * denotes no significant result.

Trained on	Tested on				
	WU	JHU	Houston	Vanderbilt	CCF
WU	—	COLA	CFOD_G	*	*
JHU	*	—	*	CFOD_G	CFOD_G
Houston	*	CFOD	—	*	COLA
Vanderbilt	CFOD, CFOD_G	*	*	—	COLA
CCF	*	*	*	CFOD	—

Three patterns emerged.

CFOD_G was the most consistently predictive of the three families, reaching significance more often than either CFOD or COLA.

Results were better under DFS than under OS. The most likely explanation is simply that the models and features are not strongly prognostic and the apparent DFS signal was partly noise. A more interesting possibility, which this data cannot distinguish, is that CFOD features are genuinely more informative about recurrence than about overall mortality. Separating those two explanations would require a larger cohort and a pre-registered comparison.

Table 3. The same framework after the endpoint was changed to overall survival.

<i>Trained on</i>	<i>Tested on</i>				
	WU	JHU	Houston	Vanderbilt	CCF
WU	—	*	*	*	*
JHU	*	—	CFOD	CFOD, CFOD_G	*
Houston	*	*	—	*	*
Vanderbilt	*	*	*	—	*
CCF	*	*	*	*	—

The stricter frameworks produced weaker results. Neither the 60:40 nor the 70:30 development split yielded significant results for any of the three feature families, and swapping the development and testing sets to enlarge development did not recover significance either. The most permissive framework produced the most significant findings, which is the ordering one would expect if the earlier results were largely artefacts of validation design.

The strongest individual results came from the JHU and Vanderbilt datasets, where CFOD_G and CFOD were most consistent, the Kaplan–Meier curves showed better separation between risk groups, and hazard ratios were more clinically meaningful.

1.5 Obstacles

Dataset volume and survival analysis runtime were the persistent practical constraints. Interpretation was the persistent conceptual one: features that were prognostic for one institution’s data frequently were not for another, with no obvious covariate explaining which. The development-testing split was the most instructive difficulty, because it demonstrated that a stricter validation framework combined with a changed endpoint could remove nearly all apparent signal from a set of results that had previously looked encouraging.

1.6 What the project established

The concrete methodological takeaway is how much study design and data collection choices move statistical outcomes. The DFS to OS comparison is a clean demonstration: the underlying patients, features and models were unchanged, and the conclusions changed substantially. In practical terms the semester also covered Kaplan–Meier estimation, log-rank testing and Cox proportional hazards models applied to real multi-institutional data, none of which was familiar beforehand. The work was done in consultation with Himanshu, German and other lab collaborators, each specialising in a different part of the pipeline.

2 Project II: Histopathology Agentic Super Classifier

2.1 Problem

Pathologists have access to thousands of computational tools and very little practical means of choosing among them. Selecting, configuring, running and interpreting the right pipeline for a given question is neither intuitive nor fast. The goal was an agent that handles selection and execution so the pathologist can concentrate on interpretation and initial diagnosis.

2.2 Architecture

Three components: the lab’s existing feature extraction tools, a natural language interface, and a results presentation layer. The agent, built on Anthropic Claude, interprets the query, determines which code to

execute, runs it and organises the output.

Five tool families were integrated, adapted from established lab pipelines: tumour-infiltrating lymphocytes, mitosis detection, tumour segmentation, collagen fibre analysis (building directly on the CFOD work above) and nuclei analysis. The structure is modular by design. Adding a tool requires registering it in the agent prompt with its preprocessing entry point, available features and a short description, which is a small enough change that a researcher can contribute a tool without touching the orchestration logic.

Selectivity was the property that received the most attention. Asked about “mitotic count and tubule formation in this H&E image,” the agent should run mitosis detection and tubule analysis and leave collagen and TIL pipelines untouched. Running everything would defeat the purpose.

The interface accepts uploaded images (patches initially, whole-slide images as the intended endpoint) and plain-language questions, then returns the script outputs including generated masks, alongside an interactive visualisation of the analysis workflow: which tools ran, in what order, and how the agent handled the query. Every processed query generates a directed graph of its own execution.

2.3 Obstacles

Each extraction tool arrived with its own environment and dependency requirements, which made integration the bulk of the effort. The agent required detailed system prompts defining its role, its available tools and its decision-making framework before tool selection became reliable. The workflow visualisation added complexity but turned out to be the most valuable single feature, because it makes the agent’s reasoning auditable rather than opaque.

2.4 Review and roadmap

The system was presented to Dr. Madabhushi for review. He identified three extensions: move from patches to full whole-slide images, incorporate tools beyond histopathology such as radiology and genomic analysis, and add retrieval of similar cases from a reference dataset. The roadmap that followed organises tools into suites, a Pathomics suite and a Radiomics suite feeding downstream Surgical and Radiation Oncology suites, so that a query can traverse modalities rather than staying inside one.

2.5 Potential applications

With further development the system could give pathologists faster access to the data primitives that feed initial diagnosis and severity assessment, including the mitotic figures, tubules and nuclei that contribute directly to Nottingham histological grade. It has an obvious secondary use in training, helping new pathologists recognise patterns in features and outcomes, particularly once similar-case retrieval is implemented. In research settings it could reduce discovery time by applying relevant pipelines across larger datasets automatically and generating structured documentation of dataset features.

3 Reflection

The two projects produced opposite kinds of understanding. CFOD survival analysis demonstrated how much effort separates a basic scientific insight from a clinically usable tool, and how easily a promising result dissolves under stricter validation. The agentic classifier demonstrated the opposite: that a substantial gain is sometimes available not from new capability but from making existing capability usable. Given how many computational pathology tools exist, it is surprising how few projects work on integrating them into something a clinician would actually reach for.

Both depended entirely on collaboration. The CFOD work drew on separate specialists for feature extraction, dataset splitting and survival interpretation; the agent was a group effort with the other interns,

Rohan and Aidan. The technical skills and the research disposition built over the semester both transfer, and I am grateful to Dr. Madabhushi and the lab for the opportunity.

Document note. The two result matrices are presented here as tables. Supporting figures held with the original working documents (Kaplan–Meier curves, the tool repository README and the tool-suite roadmap) are not reproduced.